

Customer Personality Analysis & Customer Segmentation Project

Final Project Report

AI Lab 99 Internship Program | Machine Learning / Data Analytics Track

Chapter 1 — Introduction

1.1 Project Overview

This project builds a complete, production-style machine learning application around the Customer Personality Analysis dataset: from raw-data collection through cleaning, feature engineering, unsupervised segmentation, business-insight generation, and finally a live, interactive Streamlit dashboard that serves real-time customer segment predictions.

1.2 Business Problem

The business does not have a systematic way to group its 2,237 customers into actionable segments. Without segmentation, marketing spend, discounting, and retention efforts are applied uniformly rather than targeted — wasting budget on low-response segments and under-investing in high-value customers.

1.3 Motivation

Behavioural segmentation is one of the highest-leverage applications of unsupervised machine learning in marketing analytics: it converts raw transactional data into a small number of interpretable customer personas that non-technical stakeholders can act on directly.

1.4 Objectives

- Clean and validate the raw dataset end-to-end.
- Engineer behaviourally meaningful features.
- Train and compare multiple clustering algorithms; select and validate the best one.
- Translate clusters into named, business-actionable segments with strategy recommendations.
- Ship a tested, documented, interactive dashboard application exposing all of the above.

1.5 Scope

The project covers the full pipeline from raw CSV to deployed dashboard, using a single historical snapshot of the dataset (2,237 customers). It does not include real-time data ingestion, automated retraining, or production cloud deployment — these are discussed as future improvements in Chapter 12.

Chapter 2 — Dataset

Property	Value
Source	Customer Personality Analysis dataset (Kaggle)
Raw Size	2,240 rows x 29 columns, 215 KB, CSV, UTF-8
Final Processed Size	2,237 rows x 67 columns (after cleaning, feature engineering, segmentation, strategy)

2.1 Features

The raw dataset covers customer demographics (birth year, education, marital status, income, household composition), enrollment date, purchasing behaviour across 6 product categories and 4 channels, responses to 6 marketing campaigns, complaints, and website engagement.

2.2 Data Dictionary (Key Fields)

Field	Description
Income	Annual household income
Recency	Days since the customer's last purchase
MntWines / MntFruits / MntMeatProducts / ...	Amount spent per product category (2 years)
NumWebPurchases / NumStorePurchases / NumCatalogPurchases	Purchases per channel
AcceptedCmp1-5, Response	Whether the customer accepted each of 6 campaigns
Complain	Whether the customer filed a complaint

2.3 Data Quality Issues Identified

- 24 missing Income values (~1.07%).
- 3 customers with implausible ages (>100 years).
- Inconsistent Marital_Status categories ("Alone", "Absurd", "YOLO").
- 2 constant columns (Z_CostContact, Z_Revenue) carrying no information.
- Right-skewed Income and spending distributions requiring transformation.

Chapter 3 — Data Preprocessing

3.1 Missing Values

Income (1.07% missing) was imputed using the median rather than the mean, since income is right-skewed by a small number of high earners and the median is a more robust central estimate.

3.2 Duplicates

Full-row and Customer-ID duplicate checks were run; 0 duplicate rows or IDs were found in the raw data.

3.3 Data Types

Dt_Customer was parsed from text into a proper datetime type; ID, Year_Birth, Recency and counts were cast to integer; Income to float; categorical fields to category dtype.

3.4 Outliers

Income and the six spending columns were analyzed via the IQR method. Income outliers (8 extreme-but-plausible high earners) were capped (winsorized) at the IQR upper bound; spending-column outliers were deliberately kept, since right-skewed high spending reflects genuine customer behaviour rather than data error.

3.5 Encoding

Education and Customer_Activity_Level were label-encoded (they have a natural order); Marital_Status, Preferred_Shopping_Channel, and Product_Preference were one-hot encoded (no inherent order).

3.6 Scaling

StandardScaler was selected over MinMaxScaler and RobustScaler comparisons for the final clustering feature set, since K-Means/GMM/Agglomerative clustering rely on Euclidean distance which performs best on zero-mean, unit-variance features.

Chapter 4 — Exploratory Data Analysis

4.1 Important Findings

- Customers range 30-86 years old (mean 57.1); income averages \$51,855, right-skewed.
- Wine is the highest-revenue product category across the entire customer base.
- Store purchases are the most-used channel, ahead of web and catalogue.
- The latest marketing campaign achieved a 14.9% acceptance rate — more than double any prior campaign.
- Income and Total Spend correlate strongly (Pearson $r \sim 0.80$); web-visit frequency is negatively correlated with both.

4.2 Customer Behaviour & Spending Patterns

Higher-income customers spend more overall and show stronger campaign responsiveness. Households with children spend markedly less than those without, reflecting more budget-constrained, value-oriented purchasing.

4.3 Campaign Behaviour

Only ~0.9% of customers filed a complaint; complainers show both lower income and lower spend than non-complainers, consistent with dissatisfaction concentrated among lower-value customers.

Chapter 5 — Feature Engineering

Feature	Formula	Reason
Customer_Age	Reference_Year - Year_Birth	Life-stage / demographic driver
Customer_Tenure	Reference_Date - Dt_Customer (days)	Loyalty / relationship-length signal
Total_Children	Kidhome + Teenhome	Combined dependents indicator
Family_Size	1 + partner + Total_Children	Household-scale indicator
Total_Spending	Sum of 6 Mnt* columns	Direct customer-value measure
Total_Purchases	Sum of 4 channel purchase counts	Purchase frequency
Total_Campaign_Acceptance	Sum of 6 campaign flags	Marketing receptiveness
Avg_Spending_Per_Purchase	Total_Spending / Total_Purchases	Basket-value refinement
Digital_Engagement	WebVisits + WebPurchases	Online activity score
Deal_Dependency	DealsPurchases / Total_Purchases	Price-sensitivity signal

13 engineered features in total were created (see the Module 05 Feature Dictionary for the complete set, including Preferred_Shopping_Channel, Product_Preference, and Customer_Activity_Level).

Chapter 6 — Machine Learning

6.1 Algorithms Tested

Algorithm	Silhouette	Davies-Bouldin	Clusters	Time (s)
K-Means (selected)	0.169	1.890	4	0.017
Hierarchical (Agglomerative)	0.139	1.805	4	0.166
Gaussian Mixture Model	0.048	2.918	4	0.094
DBSCAN	N/A	N/A	1 + noise	0.128

6.2 Model Selection Process

K was determined using the Elbow Method, Silhouette Score, and Davies-Bouldin Index across K=2-10. Silhouette and Davies-Bouldin both statistically favored K=2, but this was judged too coarse to produce the actionable, named personas the business needed. K=4 was chosen as a documented, business-driven override: the elbow's WCSS improvement visibly flattens from K=4 onward, and the K=4 silhouette score (~0.17) remains a reasonable, positive separation score.

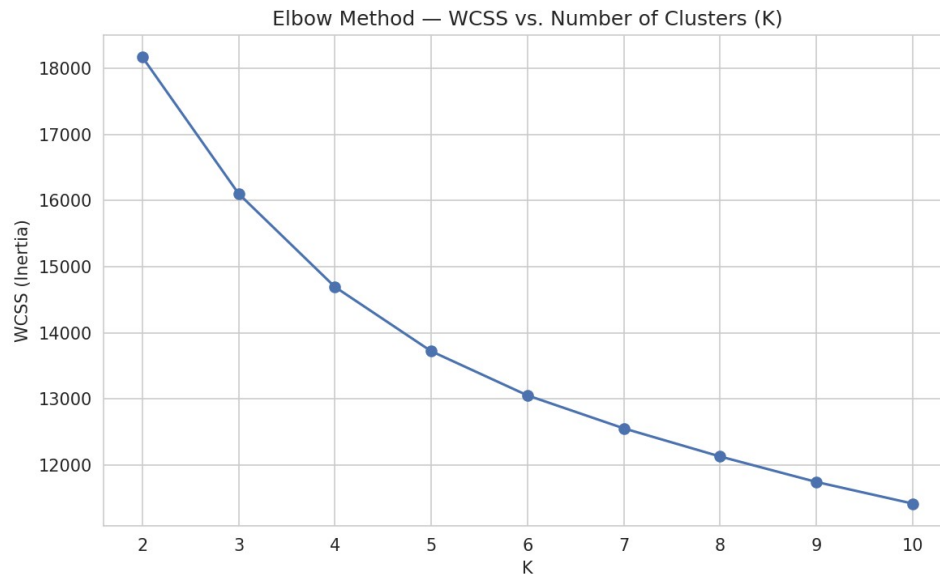


Figure 6.1 — Elbow plot used in K selection.

6.3 Final Model

K-Means (K=4, StandardScaler-normalized features) was selected as the production model based on its strong Silhouette/Davies-Bouldin scores, fastest training time, highest interpretability (centroids map directly to segment averages), and high stability (ARI > 0.95 across random seeds and bootstrap resamples).

Chapter 7 — Customer Segmentation

Segment	Size	% Share	Avg Income	Avg Spend
High-Value Customers	495	22.1%	\$76,705	\$1,430
Premium / Loyal Buyers	639	28.6%	\$59,927	\$814
Discount Seekers / Budget Customers	427	19.1%	\$42,574	\$131
New / Developing Customers	676	30.2%	\$31,891	\$105

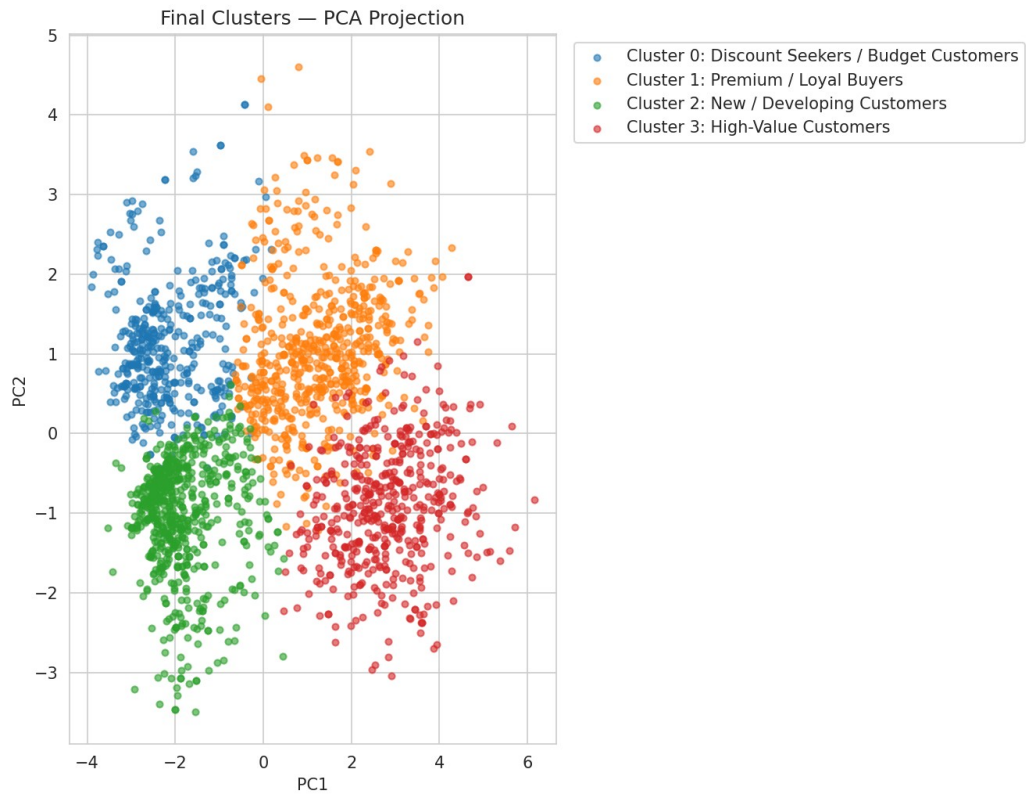


Figure 7.1 — Final K-Means clusters, PCA projection.

Chapter 8 — Business Recommendations

- **Marketing:** low-frequency VIP campaigns for High-Value Customers; increased campaign investment for the highly-engaged Premium/Loyal segment; value-led SMS/Email for Discount Seekers; structured onboarding for New/Developing Customers.
- **Product:** Wine and Meat are the strongest primary/secondary categories across all segments; Gold products are the consistent upsell path.
- **Discount strategy:** 5-10% for High-Value (premium pricing candidate) up to 15-25% for Discount Seekers (highest price sensitivity).
- **Retention:** tiered loyalty — VIP, points-based, cumulative-discount, and onboarding-rewards tiers matched to each segment's value.
- **Cross-selling / Upselling:** Gold/Premium-tier upgrades recommended universally; Meat and Fish as the primary cross-sell targets.
- **Reactivation:** 1,103 customers across all segments (Recency in the top quartile) are flagged for win-back campaigns, timed from immediate (High-Value) to 30-45 days (New/Developing).

Business Insights Dashboard — Customer Segmentation

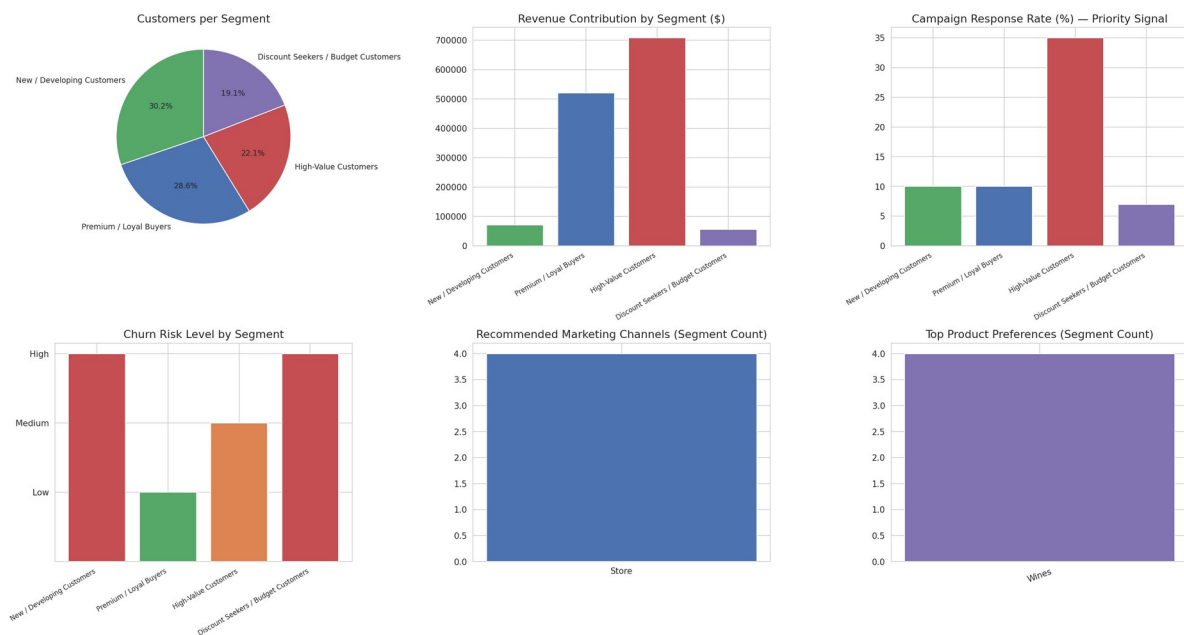


Figure 8.1 — Business Insights Dashboard summarizing all four segments.

Chapter 9 — Dashboard

9.1 Architecture

The Streamlit dashboard (app/app.py, app/dashboard.py, app/components/) exposes 10 sections via a sidebar navigation: Executive Overview, Demographics, Spending, Product, Channel, Campaign Response, Segments, Segment Comparison, Individual Customer Profile, and Marketing Recommendations. A shared filter panel (segment, age, income, education, marital status, product, channel, campaign response) drives every section live.

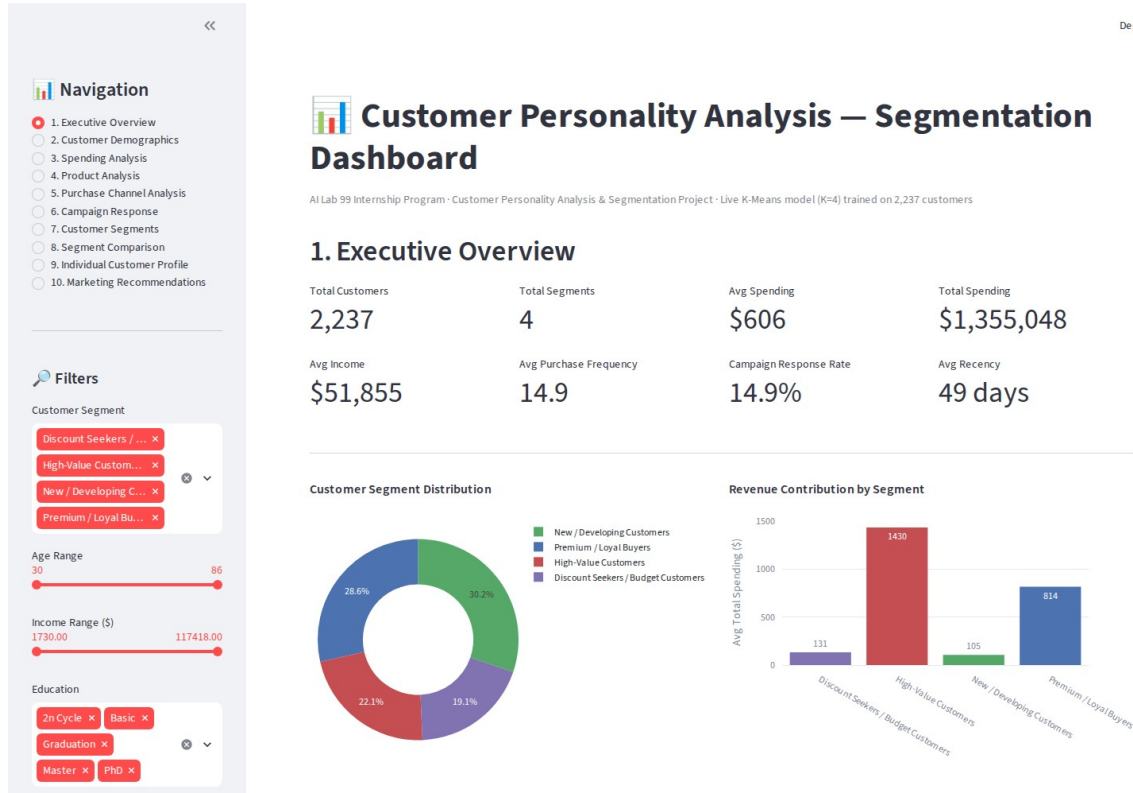
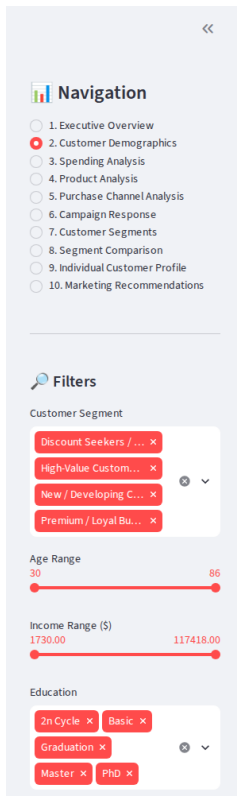


Figure 9.1 — Executive Overview section.

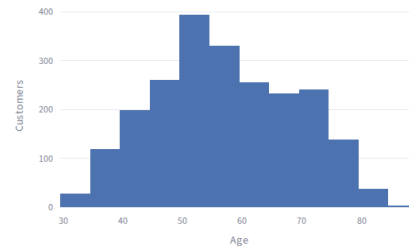


Customer Personality Analysis — Segmentation Dashboard

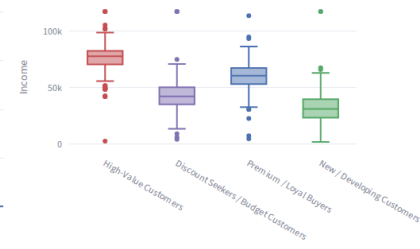
AI Lab 99 Internship Program · Customer Personality Analysis & Segmentation Project · Live K-Means model (K=4) trained on 2,237 customers

2. Customer Demographics

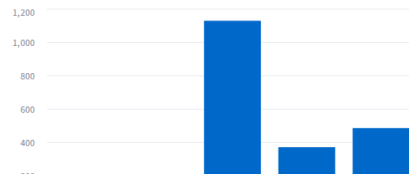
Age Distribution



Income Distribution by Segment



Education Level



Marital Status

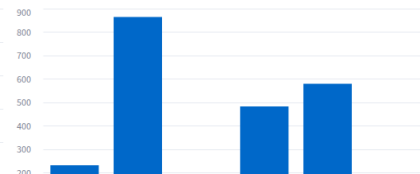
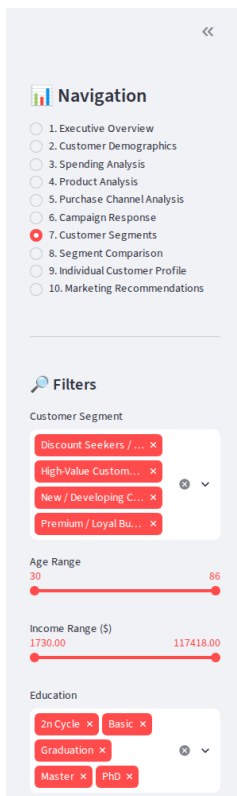


Figure 9.2 — Customer Demographics section.



Customer Personality Analysis — Segmentation Dashboard

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7. Customer Segments

Segment_Name	Customers	Avg_Income	Avg_Spending	Avg_Recency	Percentage
Discount Seekers / Budget Customers	427	42574.2	131.3	50.2	19.1
High-Value Customers	495	76705	1430.1	50.1	22.1
New / Developing Customers	676	31890.6	104.9	49	30.2
Premium / Loyal Buyers	639	59926.5	814	47.7	28.6

Segment Characteristics

- > Discount Seekers / Budget Customers (427 customers)
- > High-Value Customers (495 customers)
- > New / Developing Customers (676 customers)
- > Premium / Loyal Buyers (639 customers)

Figure 9.3 — Customer Segments section.

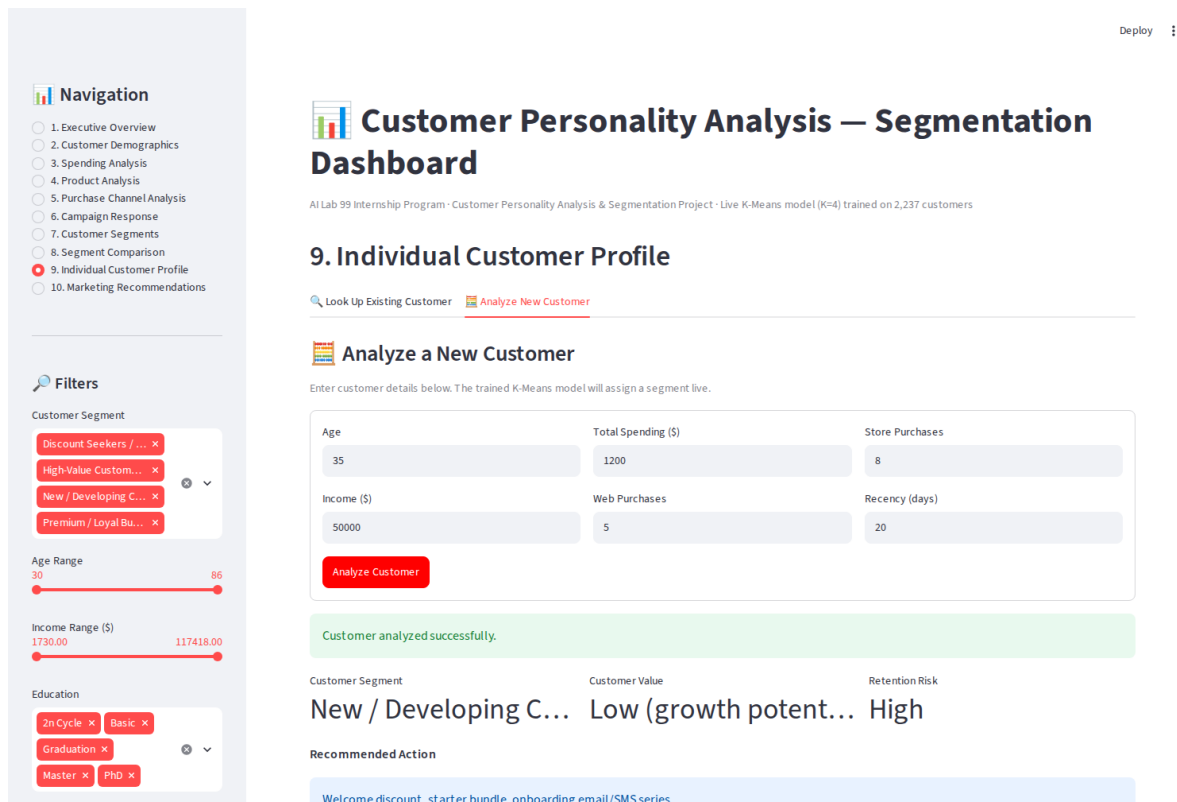


Figure 9.4 — Live individual-customer prediction using the trained K-Means model.

9.2 Features

- 8 interactive filters that update all charts/tables live.
- Customer ID lookup against the full dataset.
- A live prediction form that runs a NEW customer through the exact same feature-engineering and scaling pipeline used in training.
- Two-segment comparison with a radar chart and downloadable CSV.
- Clear, non-crashing error handling for invalid input.

Chapter 10 — Testing & QA

40 automated pytest tests were written across 4 suites (data validation, preprocessing, feature engineering, model) — all 40 pass. Additional manual testing covered the dashboard UI, user-acceptance scenarios, and performance.

Test Suite	Tests	Result
Data Validation	9	9/9 Pass
Preprocessing (input validation)	12	12/12 Pass
Feature Engineering	12	12/12 Pass
Model / Prediction	7	7/7 Pass

3 defects were found and fixed during development (see reports/21_defect_report.xlsx): a minor sklearn feature-name warning, a cluster-naming inconsistency between the Module 06 script and notebook, and a text-overflow issue in the Module 08 HTML dashboard. Regression testing confirmed all fixes did not introduce new issues.

User Acceptance Testing found the dashboard easy to navigate, filter, and interpret, with no failures reported; the full UAT log is in reports/20_uat_report.xlsx.

Chapter 11 — Limitations

- Dataset limitations: a single historical snapshot (2,237 customers); no time-series/behavioural-trend data; some fields (e.g. Income) contain repeated values from median imputation and natural rounding in the source survey.
- Model limitations: K-Means assumes roughly spherical, similarly-sized clusters; the chosen K=4 was a business-driven override of the statistically-optimal K=2, which is a defensible but subjective judgment call.
- Business limitations: recommendations are grounded in historical averages and have not yet been validated against live campaign outcomes.
- Dashboard limitations: runs against a static CSV snapshot rather than a live database connection; no user authentication or multi-user support.

Chapter 12 — Future Improvements

- Real-time customer data integration via a database or API connection instead of a static CSV.
- Automated model retraining on a schedule as new customer data arrives.
- A full recommendation engine (item-level, not just category-level) using collaborative filtering.
- Customer lifetime value (CLV) prediction using regression/survival models rather than the current simple 3-year spend proxy.
- Churn prediction as a supervised complement to the unsupervised segmentation.
- Cloud deployment (e.g. Streamlit Community Cloud, AWS, or Azure) with proper authentication.
- Real-time campaign integration so marketing actions triggered from the dashboard flow directly into a CRM/ESP.

Chapter 13 — Conclusion

This project delivered a complete, working machine learning application: a cleaned and validated dataset, 13 engineered behavioural features, a K-Means segmentation model selected from four candidate algorithms and validated for stability, four named and profiled customer segments, a full set of business and marketing recommendations grounded in the data, and a tested, documented, interactive Streamlit dashboard that serves live predictions using the exact same pipeline logic used in training. The analysis shows a classic revenue-concentration pattern — 22.1% of customers (High-Value) generate 52.2% of revenue — while the largest segment by count (New/Developing, 30.2%) represents the primary growth opportunity. Executing the accompanying marketing strategy matrix is expected to improve retention, lift customer lifetime value, and make marketing spend materially more efficient across all four segments.